MACHINE LEARNING FOR CREDIT RISK MODELING

A Comparative Study of Traditional and Explainable AI Approaches

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Executive Summary

The research question, approach, result and practical meaning

Problem

Credit risk models must detect rare defaults while remaining explainable and defensible.

Approach

Compare Logistic Regression, Random Forest, XGBoost and LightGBM; use SMOTE and SHAP.

Finding

LightGBM was selected for the most useful minority-class operating point despite RF's slightly higher AUC.

Outcome

A transparent, leakage-aware framework suited to triage—not autonomous lending.

3.4%
exploratory threshold

7.1%
precision

32.1%
recall

0.659
LightGBM ROC AUC

Research Objectives

1. **Compare Logistic Regression with Random Forest, XGBoost and LightGBM.**
2. Address the 1.8% default-rate imbalance using SMOTE.
3. Demonstrate transparency using SHAP values.
4. Conduct a preliminary fairness analysis using income as a proxy.

Data & Problem Structure

OpenIntro Loans Full Schema

10,000	55	1.8%	113
loans	raw variables	default rate	encoded predictors

Key financial variables

- Annual income
- Debt-to-income ratio
- Loan amount
- Interest rate
- Loan grade / term

Target

Binary classification
0 = Fully Paid / Current
1 = Default / Charged Off

Methodology Pipeline

Leakage prevention is central to the study

1 **Clean & validate** → 2 **Drop post-origination leakage** → 3 **70/30 stratified split** → 4 **SMOTE on training only** → 5 **Train 4 models** → 6 **SHAP + fairness analysis**

Critical validation insight

Including post-origination repayment artifacts inflated performance to ROC AUC 0.83 and precision 55.6%. Removing them produced the honest at-origination baseline: ROC AUC 0.66 and optimized precision 7.1%.

Why Class Imbalance Matters

Class distribution



1.8% default

Accuracy paradox

A naive all-good classifier reaches ~98% accuracy while detecting essentially no defaults.

SMOTE response

SMOTE was applied only to the training partition to create a balanced 50/50 learning set without contaminating the hold-out test set.

Model Comparison

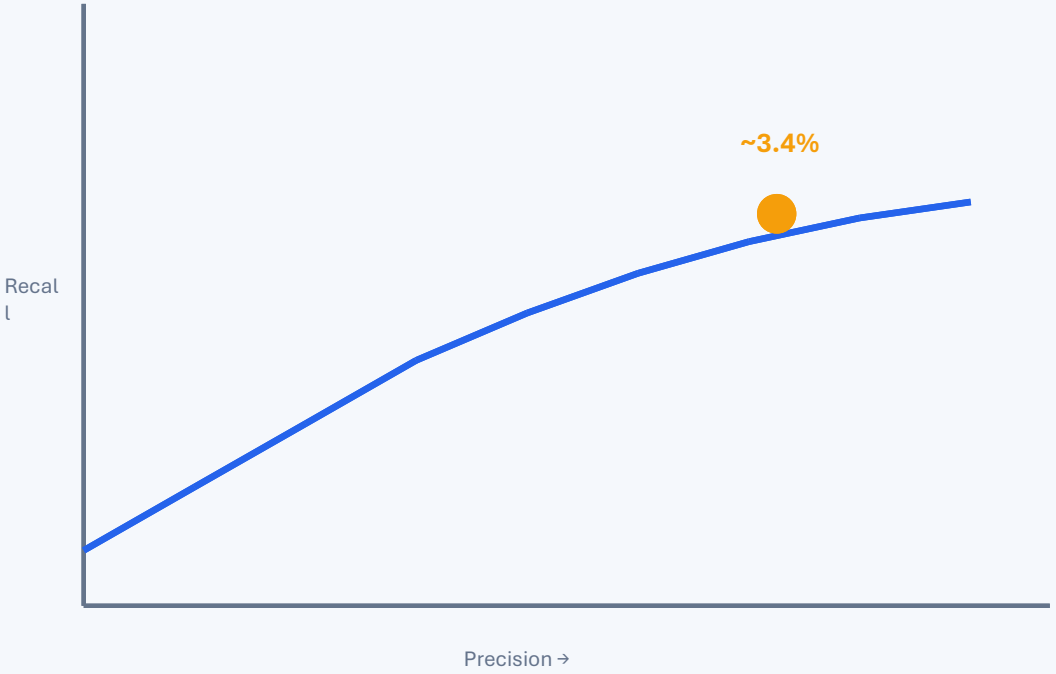
Standard 0.50 threshold metrics from the thesis

Model	ROC AUC	Precision	Recall	F1
Random Forest	0.670	0.000	0.000	0.000
LightGBM	**0.659**	**0.250**	**0.019**	**0.035**
XGBoost	0.644	0.200	0.019	0.034
Logistic Regression	0.575	0.023	0.472	0.044

Interpretation Random Forest has the highest AUC, but at 0.50 it collapses to the majority class. LightGBM was therefore chosen for threshold optimization and minority-class detection.

Decision Threshold Optimization

The 3.4% operating point is exploratory—not a final deployment threshold



Precision

7.1%

Recall

32.1%

Caveat

Threshold tuned on the hold-out test set; metrics are optimistic.

Explainable AI with SHAP

From global importance to borrower-level reasoning

1 • Global drivers

Loan grade
Interest rate
Income verification
Loan term
Debt-to-income

2 • Direction matters

SHAP shows not only which features matter, but whether each feature pushes the prediction toward higher or lower default risk.

3 • Local explanation

For an individual borrower, SHAP decomposes the prediction into risk amplifiers and mitigating factors—supporting adverse-action rationale.

Key message: a complex model can be made inspectable at both portfolio and applicant level.

Borrower-Level Case Study

Why local explanations matter

Example model output

≈ 1.7%

uncalibrated probability for the illustrated borrower

Risk amplifier

grade_D pushed the score upward.

Mitigator

Payment history and other features reduced risk.

Important nuance

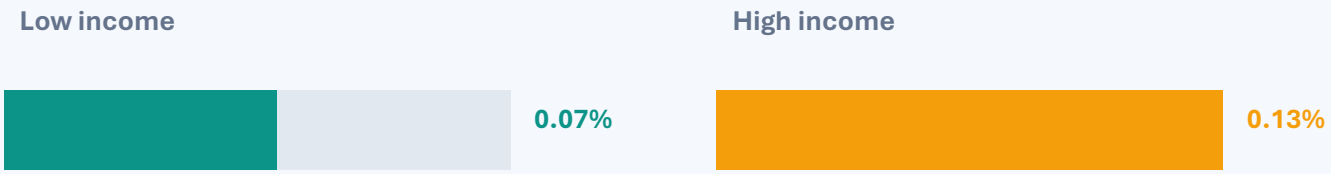
A 60-month term increased risk globally but reduced risk for this borrower locally. The divergence illustrates nonlinear interaction effects captured by LightGBM.

SHAP therefore complements, rather than replaces, underwriting judgment.

Fairness & Compliance

Preliminary evidence only

False Positive Rate



Observed result

No evidence of disparate impact under the evaluated FPR-parity metric; low-income FPR was lower (0.07% vs 0.13%).

Limitation

The dataset lacks protected demographic attributes and contains very few default events; this is not a definitive regulatory audit.

Stress Testing & Production Implications

Adversarial stress

50 extreme profiles stressed interest rate and DTI. Outputs remained bounded and numerically stable.

But no extrapolation

Maximum stressed PD was 0.422 vs 0.675 in-distribution—showing the tree model can understate novel tail risk.

Governance

Production use requires drift monitoring, validation, optimized SHAP inference and human oversight.

Recommended role

TRIAGE & PRIORITIZATION

Use the PD score to prioritize manual review, not as an autonomous approval/denial engine.

Limitations & Threats to Validity

- Only 10,000 observations; 113 encoded predictors increase overfitting risk.
- Threshold selection used the hold-out test set, introducing optimistic bias.
- Recall remains 32.1% at the exploratory operating point.
- Fairness analysis relies on income proxies because protected demographic fields are absent.
- No statistical significance tests such as McNemar's test or bootstrapped confidence intervals were performed.
- Stress testing shows tree-based models may understate genuinely out-of-distribution risk.

These limitations are part of the contribution: the thesis explicitly distinguishes promising evidence from deployment-grade validation.

Future Research & Deployment Roadmap

01

Two-stage PD → LGD/EAD → Expected Loss

02

Cost-sensitive threshold optimization

03

Full demographic fairness validation

04

Production deployment + drift monitoring

05

Larger, temporal and external validation

Conclusion

A constrained LightGBM + SHAP framework can improve minority-class credit-risk detection while preserving interpretability.

Predictive

LightGBM provided a useful operating point under severe imbalance.

Transparent

SHAP connected model outputs to portfolio and borrower-level reasoning.

Responsible

Leakage prevention, fairness caveats and stress testing were treated as first-class requirements.